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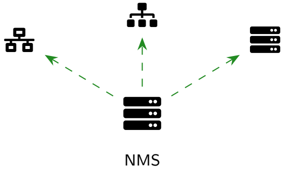
RCTC

Introduction to Networking



Module 8 Learning Objectives

1. Documentation & Policies

- Configuration management
- Change management processes
- Physical & logical diagrams
- Asset and lifecycle management

2. Discovery & Monitoring

- Network discovery (nmap)
- Performance monitoring
- Availability and configuration checks

3. SNMP & Event Management

- SNMP agents, versions, security
- Syslog and centralized logging
- SIEM and event correlation

4. Traffic Analysis

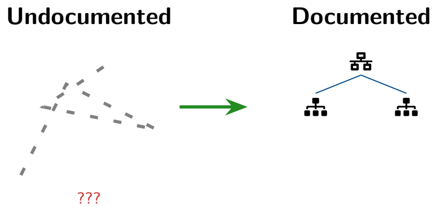
- Packet capture (tcpdump, Wireshark)
- Flow data (NetFlow, sFlow)
- QoS and traffic shaping

Section 8.1: Why Documentation Matters

The Swamp Network Problem

Without documentation, networks become **unmaintainable swamps**:

- No one knows what's connected where
- Changes break unknown dependencies
- Troubleshooting takes hours instead of minutes
- New staff can't understand the environment



Key Takeaway

Good documentation is the difference between a **manageable network** and a **chaotic swamp**!

Documentation Types

Configuration management, change management, asset inventory, network diagrams, IPAM

Configuration Management

What is Configuration Management?

The practice of maintaining **consistent, documented device configurations** across your network infrastructure.

- Baseline configurations (templates)
- Version control for changes
- Automated compliance checking
- Rollback capabilities

Configuration Management Elements

Element	**Purpose**
Baseline	Standard reference
Version Control	Track all changes
Validation	Check compliance
Backup	Enable recovery
Templates	Consistent deploys

Why It Matters

Configuration drift causes outages! Consistent configs mean predictable behavior.

Common Tools

RANCID, **Oxidized**, SolarWinds NCM, Ansible, Git for network configs

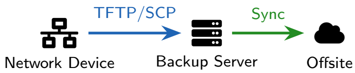
Network Device Backup Management

What to Backup

- **Startup-config**: Saved configuration
- **Running-config**: Active configuration
- **Firmware/IOS images**: Operating system
- **Certificates & keys**: Security material
- **License files**: Entitlements

The 3-2-1 Rule

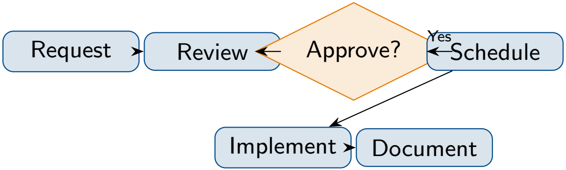
3 copies of data, on **2** different media types, with **1** copy offsite



Backup Schedule

Frequency	What
On change	Running-config
Daily	All configs
Weekly	Firmware images

Change Management Process



CAB = Change Advisory Board

Team that reviews and approves changes:

- Assesses risk and impact
- Schedules maintenance windows
- Ensures rollback plans exist

Change Types

Type	Example	Approval
Standard	Password reset	Pre-approved
Normal	VLAN addition	CAB review
Emergency	Security patch	Expedited

Key Principle

No undocumented changes! Every change needs a ticket, approval, and verification.

Case Study: Donkey's Unauthorized Network Change

"I'm Making Waffles... and WiFi!"

Donkey decides to "help" by adding a rogue wireless access point to the Swamp Castle network so visitors can connect more easily. He plugs it directly into a core switch port.

The next morning, Lord Farquaad's security team detects multiple unknown devices on the network. Investigation reveals the unsecured AP has been broadcasting "SwampCastle-Guest" with no password, and several unauthorized users have connected—including someone running network scans.

Current situation: Rogue AP on VLAN 1 (management VLAN), no encryption, unknown devices accessing internal resources.

Review Questions

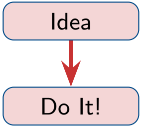
1. What change management steps did Donkey skip?
2. What documentation should exist for wireless additions?
3. How could this security breach have been prevented?

Case Study Solution: Donkey's Unauthorized Change

Solution: "That'll Do, Donkey—After Proper Approval!"

1. Donkey skipped **all** steps: no request, no review, no approval, no documentation
2. Required docs: Change request, security review, network diagram update, IPAM entry
3. Prevention: **Port security**, **802.1X**, **change management policy training**

Donkey's Way



Correct Way

Key Lesson

Even **helpful** changes can cause security breaches. Change management exists to protect the network—not to slow people down!

Asset Inventory Documentation

What to Track

- **Hardware**: Routers, switches, servers, APs
- **Software**: OS versions, licenses, patches
- **Location**: Rack, room, building, site
- **Ownership**: Department, cost center
- **Lifecycle**: Purchase, warranty, EOL dates

Why Track Assets?

Security (know what's on network), compliance (license audits), planning (budget/refresh), troubleshooting

Asset Inventory Fields

Field	Example
Asset Tag	SW-CORE-001
Hostname	swamp-sw-01
Serial Number	FTX1234ABCD
MAC Address	AA:BB:CC:DD:EE:FF
IP Address	10.1.1.2
Location	Swamp DC, Rack 3
Purchase Date	2024-01-15

Lifecycle Management & Decommissioning

Key Lifecycle Terms

- **EoS** (End of Sale): No longer sold
- **EoSW** (End of Software): No new features
- **EoL** (End of Life): No support/patches
- **Decommission**: Proper removal

Decommissioning Checklist

1. Backup final configuration
2. Document current connections
3. Notify affected users
4. **Wipe all data securely**
5. Revoke certificates/keys
6. Update inventory/diagrams

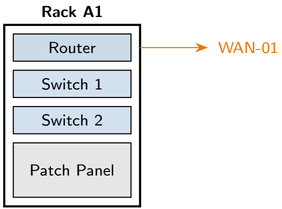
Security Warning

Old devices contain configs, credentials, and keys. **Always sanitize before disposal!**

Physical Network Diagrams

What Physical Diagrams Show

- **Actual device locations** (racks, rooms)
- **Cable runs** with labels and types
- **Port assignments** (Gi0/1 to Patch 24)
- **Rack elevations** and power connections
- Physical redundancy paths



Use Cases

Technician dispatches, cable tracing, capacity planning, disaster recovery

Diagram Elements

Device icons, cable lines, port labels, physical location markers

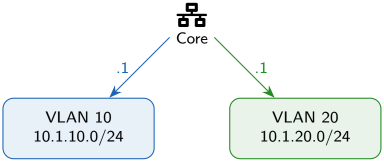
Logical Network Diagrams

What Logical Diagrams Show

- **IP addressing** and subnets
- **VLAN assignments**
- **Routing relationships**
- **Logical groupings** (not physical)
- Layer 3 topology

Physical vs Logical

Physical	**Logical**
Where things are	How things connect
Cable paths	IP subnets
Rack locations	VLAN boundaries



Both Are Needed!

Physical diagrams for hands-on work; logical diagrams for troubleshooting traffic flow

IP Address Management (IPAM)

IPAM Purpose

- **Track** all IP allocations
- **Prevent** address conflicts
- **Plan** for growth
- **Document** assignments
- **Integrate** with DNS/DHCP

IP Allocation Documentation

Subnet	VLAN	Use	GW
10.1.10.0/24	10	Users	.1
10.1.20.0/24	20	Servers	.1
10.1.30.0/24	30	Guest	.1
10.1.99.0/24	99	Mgmt	.1

IPAM Tools

phpIPAM (free), **NetBox** (free), Infoblox, SolarWinds IPAM, BlueCat

Avoid These Problems

Duplicate IP addresses, overlapping subnets, exhausted DHCP pools, undocumented static IPs

Common Agreements: SLA, MOU, NDA

Agreement Types

Type	Binding?	Purpose	Example
SLA	Yes	Service guarantees	99.9% uptime
MOU	Sometimes	Partnership terms	Data sharing
NDA	Yes	Confidentiality	Trade secrets
SOW	Yes	Work scope	Project tasks

SLA Key Metrics

Uptime: 99.9% = 8.76 hrs/year downtime; **Response time**: How fast to acknowledge; **Resolution time**: How fast to fix

SLA = Service Level Agreement

Contract defining service expectations, metrics, and consequences.

MOU = Memorandum of Understanding

Less formal agreement outlining partnership terms.

NDA = Non-Disclosure Agreement

Legal contract protecting confidential information.

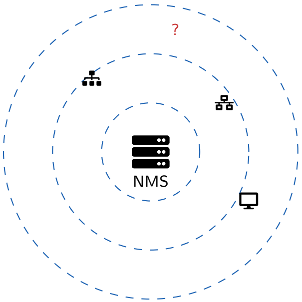
Section 8.2: Host Discovery & Monitoring

Why Discover & Monitor?

- **Security**: Find rogue devices
- **Inventory**: Validate asset lists
- **Troubleshooting**: Identify issues
- **Capacity**: Plan for growth
- **Compliance**: Audit readiness

Key Question

"What's on your network?" If you can't answer confidently, you have a visibility problem!



Discovery Protocols

ICMP (ping), ARP, SNMP, CDP/LLDP, DNS, NetBIOS

Network Discovery Methods

Active Discovery

Sends probes to find devices:

- Ping sweeps (ICMP)
- Port scans (TCP/UDP)
- SNMP queries
- ARP requests

Pro: Comprehensive results
Con: Generates traffic, may trigger alerts

Passive Discovery

Listens to existing traffic:

- ARP table analysis
- Switch MAC tables
- Traffic sniffing
- NetFlow/sFlow data

Pro: Stealthy, no extra traffic
Con: Only sees active devices

Best Practice

Use **both methods**: Active for comprehensive scans, passive for continuous monitoring

Nmap — The Network Mapper

What is Nmap?

Open-source **network scanner** for:

- Host discovery
- Port scanning
- Service/version detection
- OS fingerprinting
- Vulnerability scanning

Legal Warning!

Only scan networks you **own** or have **written permission** to scan. Unauthorized scanning is illegal!

Common Nmap Commands

```
# Ping sweep (host discovery)
nmap -sn 192.168.1.0/24
# SYN scan (stealth)
nmap -sS 192.168.1.1
# Version detection
nmap -sV 192.168.1.1
# OS detection
nmap -O 192.168.1.1
# Aggressive scan
nmap -A 192.168.1.1
```

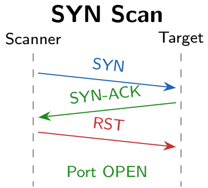
Nmap Port Scanning Techniques

Scan Types Comparison

Scan	Flag	Method	Stealth
TCP Connect	-sT	Full handshake	Low
SYN Scan	-sS	Half-open	Medium
UDP Scan	-sU	UDP probes	Medium
ACK Scan	-sA	Firewall map	High
FIN Scan	-sF	FIN flag only	High

SYN Scan (Most Common)

Default for root/admin. Sends SYN, receives SYN-ACK (open) or RST (closed), then sends RST. Never completes connection!



Case Study: Puss in Boots' Network Reconnaissance

"Fear Me, If You Dare!"

Puss in Boots has been hired as a security consultant to assess the Swamp Kingdom network. Shrek wants to know what devices are on the network and what services are exposed before the annual security audit.

Network information:

- IP range: 10.0.0.0/16 (approximately 500 devices expected)
- Business hours: 9 AM – 5 PM (critical operations)
- Shrek's requirement: "Don't break anything!"
- Deadline: Full report needed within 48 hours

Puss needs to discover all hosts, identify open ports, and detect running services without disrupting the kingdom's operations.

Review Questions

1. What scan type for initial discovery? 2. How to minimize business impact? 3. What to document?

Case Study Solution: Puss in Boots' Reconnaissance

Solution: "I Am Not a Kitten—I Am a Professional!"

1. **Phase 1**: Ping sweep (-sn) during business hours—low impact discovery
2. **Phase 2**: SYN scan (-sS) after hours—detailed port scanning
3. **Phase 3**: Version detection (-sV) on critical services only

Recommended Commands

```
# Phase 1:  Discovery (daytime)
nmap -sn 10.0.0.0/16 -oN hosts.txt
# Phase 2:  Port scan (night)
nmap -sS -iL hosts.txt -oN ports.txt
# Phase 3:  Services (targeted)
nmap -sV -p 22,80,443 -iL critical.txt
```

Key Lesson

Professional scanning is **staged and scheduled**. Balance thoroughness with business impact. Always document methodology and findings!

Discovery Protocols: CDP and LLDP

CDP (Cisco Discovery Protocol)

- **Cisco proprietary**
- Layer 2 multicast
- 60-second advertisements
- Shares: hostname, IP, platform, port ID

```
show cdp neighbors detail
```

LLDP (Link Layer Discovery Protocol)

- **IEEE 802.1AB** (vendor-neutral)
- Works across vendors
- Similar info to CDP

```
show lldp neighbors detail
```

Security Warning

CDP/LLDP reveal network info to anyone connected. **Disable on edge ports** (user-facing, untrusted connections)!

Performance Monitoring

What to Monitor

- **Bandwidth**: Link utilization %
- **Latency**: Round-trip time (ms)
- **Jitter**: Latency variation
- **Packet Loss**: % dropped packets
- **CPU**: Processor utilization
- **Memory**: RAM usage

Monitoring Tools

PRTG, Nagios, Zabbix, LibreNMS, SolarWinds, Datadog

Performance Thresholds

Metric	**OK**	**Warn**	**Crit**
CPU	<70%	70–85%	>85%
Memory	<75%	75–90%	>90%
Bandwidth	<60%	60–80%	>80%
Latency	<50ms	50–100	>100

Baseline First!

Establish **normal** performance before setting alerts. What's "high" varies by environment.

Availability Monitoring

Availability = Uptime

Measuring whether devices and services are **reachable and responding**:

- Device reachability (is it up?)
- Service availability (is it working?)
- Response time (how fast?)

Monitoring Methods

Method	Tests	Freq
ICMP Ping	Reachability	30–60s
TCP Check	Port response	1–5 min
HTTP Check	Web service	1–5 min
Synthetic	Full transaction	5–15 min

Don't Just Ping!

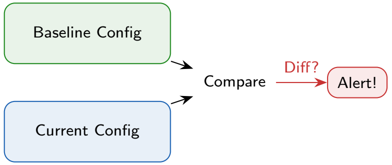
A device can respond to ping but have **failed services**. Monitor what users actually use!

Configuration Monitoring

Purpose

Detect and alert on configuration changes:

- **Unauthorized changes** (security)
- **Drift from baseline** (compliance)
- **Change verification** (operations)
- **Audit trail** (governance)



Config Monitoring Tools

RANCID, **Oxidized**, SolarWinds NCM, Cisco Prime, Ansible

What Triggers Alerts

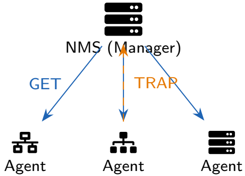
Any config change, specific keywords (passwords, ACLs), deviation from template, unauthorized user

Section 8.3: SNMP — Simple Network Management Protocol

What is SNMP?

The **standard protocol** for network device monitoring and management:

- Query device statistics
- Receive alerts (traps)
- Configure devices remotely
- Works across vendors



Key SNMP Terms

Agent: Software on managed device; **Manager/NMS**: Central monitoring station; **MIB**: Management Information Base; **OID**: Object Identifier; **Community**: Authentication string

SNMP Ports

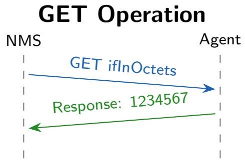
UDP 161: Agent (queries); **UDP 162**: Manager (traps)

SNMP Operations

Operation	Direction	Purpose
GET	Mgr to Agent	Read value
GET-NEXT	Mgr to Agent	Walk MIB
GET-BULK	Mgr to Agent	Read many
SET	Mgr to Agent	Write value
TRAP	Agent to Mgr	Alert (async)
INFORM	Agent to Mgr	Ack'd trap

Polling vs Traps

Polling: NMS regularly queries (GET)
Traps: Agent sends alerts immediately



Common OIDs

```
.1.3.6.1.2.1.1.1 = sysDescr
.1.3.6.1.2.1.1.3 = sysUpTime
.1.3.6.1.2.1.2.2 = ifTable
```

SNMP Versions and Security

SNMP Version Comparison

Feature	v1	v2c	v3
Authentication	Community	Community	User/Pass
Encryption	None	None	AES/DES
Integrity	None	None	MD5/SHA
GET-BULK	No	Yes	Yes
Security	Low	Low	High

Critical Security Issue!

SNMPv1/v2c send community strings in **cleartext**! Anyone sniffing traffic can capture them.

SNMPv3 Security Levels

- **noAuthNoPriv**: Username only
- **authNoPriv**: + Authentication
- **authPriv**: + Encryption

Exam Tip

SNMPv3 = **A**uthentication + **E**ncryption + **I**ntegrity. Always prefer v3 for production!

Case Study: Lord Farquaad's SNMP Disaster

"Some of You May Be Compromised..."

Lord Farquaad's IT team configured SNMP on all Duloc network devices using community strings "public" (read-only) and "private" (read-write). They thought these were just examples from the documentation.

A security audit reveals:

- Attackers have been querying device configs for weeks
- Someone changed the ACLs on the border router
- Wireshark capture shows SNMP traffic with "public" in cleartext
- All 50 network devices use identical community strings

Evidence: `snmpwalk -v2c -c public 10.0.0.1`

Review Questions

1. What SNMP vulnerabilities were exploited? 2. What version should they migrate to? 3. What immediate steps are needed?

Case Study Solution: Lord Farquaad's SNMP Disaster

Solution: "This Is Not the Fairy Tale I Wanted!"

1. **Vulnerabilities**: Default community strings + cleartext transmission (v2c) + RW access
2. **Migration**: Upgrade to **SNMPv3** with authPriv (authentication + encryption)
3. **Immediate**: Change all communities, implement ACLs, audit all configs, disable write where not needed

Before

v2c
"public"
Cleartext

→

After

v3 authPriv
SHA + AES
Encrypted

Key Lessons

Never use default credentials; v1/v2c = cleartext = vulnerable; limit SNMP access with ACLs; disable write access unless needed

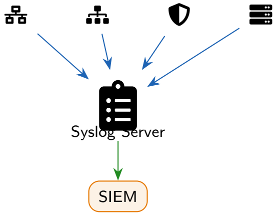
Section 8.4: Event Management & Logging

Why Logging Matters

- **Troubleshooting**: What happened when?
- **Security**: Detect breaches and attacks
- **Compliance**: Audit requirements
- **Capacity**: Trend analysis
- **Accountability**: Who did what?

Key Principle

If it's not logged, it didn't happen! Centralize logs for correlation and retention.



Log Flow

Devices → Syslog Server → SIEM → Alerts/Dashboards

Syslog Protocol and Severity Levels

Syslog Basics

- Standard logging protocol
- **UDP 514** (traditional)
- **TCP 514** (reliable)
- **TCP 6514** (TLS encrypted)

Syslog Message Format

```
<priority>timestamp host message
<134>Jan 25 10:30:15 router1
%LINK-3-UPDOWN: Interface
Gi0/1 changed state to down
```

Severity Levels (Memorize!)

Level	Name	Description
0	Emergency	System unusable
1	Alert	Immediate action
2	Critical	Critical condition
3	Error	Error condition
4	Warning	Warning condition
5	Notice	Normal but notable
6	Info	Informational
7	Debug	Debug messages

SIEM — Security Information and Event Management

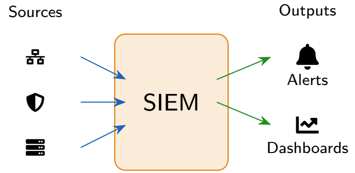
What is SIEM?

Central platform that:

- **Aggregates** logs from all sources
- **Normalizes** different log formats
- **Correlates** events across systems
- **Analyzes** for patterns/threats
- **Alerts** on suspicious activity
- **Reports** for compliance

Popular SIEM Platforms

Splunk, IBM QRadar, Microsoft Sentinel, Elastic SIEM, AlienVault



Correlation Example

Failed logins (Server) + Port scan (Firewall) + New admin (AD) = Possible breach!

Log Reviews and Alert Prioritization

Alert Priority Matrix

Priority	Response	Example
P1 Critical	15 min	Security breach
P2 High	1 hour	Service outage
P3 Medium	4 hours	Degraded perf
P4 Low	Next day	Minor warning

Alert Fatigue is Real!

Too many alerts = ignored alerts. Tune thresholds to reduce noise:

- Suppress known false positives
- Group related events
- Set appropriate thresholds
- Regularly review and tune rules

Review Schedule

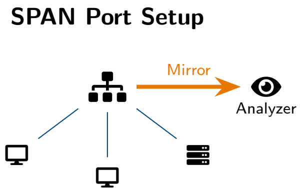
Real-time: P1/P2 alerts; **Daily**: Security events; **Weekly**: Trend analysis; **Monthly**: Compliance reports

Good Alert Design

Actionable: Clear what to do; **Contextual**: Includes relevant info; **Prioritized**: Severity is accurate

Why Capture Packets?

- **Troubleshooting**: See actual traffic
- **Security**: Detect malicious activity
- **Performance**: Analyze latency issues
- **Learning**: Understand protocols
- **Forensics**: Incident investigation



Capture Methods

SPAN/Mirror Port: Switch copies traffic; **Network TAP**: Hardware inline device; **Host-based**: Capture on endpoint

Key Tools

tcpdump: CLI packet capture; **Wireshark**: GUI protocol analyzer; **tshark**: Wireshark CLI version

tcpdump and Wireshark

tcpdump (Command Line)

```
# Capture on interface
tcpdump -i eth0
# Write to file
tcpdump -w capture.pcap
# Filter by port
tcpdump port 80
# Filter by host
tcpdump host 10.0.0.1
# Combine filters
tcpdump -i eth0 host 10.0.0.1 and port 443
```

Wireshark (GUI)

Full protocol analyzer with packet list, details, bytes views; follow TCP/UDP streams; expert analysis

Wireshark Display Filters

```
ip.addr == 10.0.0.1
tcp.port == 443
http.request
dns.qry.name contains "google"
tcp.flags.syn == 1
```

Capture vs Display Filters

Capture filters: Applied during capture (tcpdump syntax) — `host 10.0.0.1`
Display filters: Applied after capture (Wireshark syntax) — `ip.addr == 10.0.0.1`

Case Study: Princess Fiona's Network Mystery

"By Day One Way, By Night Another..."

Princess Fiona notices the Far Far Away Castle network becomes extremely slow every afternoon around 3 PM. Users complain about timeouts, dropped connections, and slow file transfers.

Basic monitoring shows:

- Core switch uplink hits 95% utilization at 3 PM
- CPU and memory on all devices are normal
- No errors or discards on interfaces
- Problem resolves itself by 5 PM

Available tools: Wireshark laptop, SPAN port access on core switch

Fiona needs to identify what's consuming all the bandwidth!

Review Questions

1. How should Fiona configure packet capture? 2. What Wireshark features help? 3. What metrics to analyze?

Case Study Solution: Princess Fiona's Network Mystery

Solution: "Better Out Than In!"

1. Configure SPAN to mirror uplink traffic to analyzer laptop during peak time
2. Use Wireshark **Statistics** → **Conversations** and **Endpoints** to find top talkers
3. Analyze: bytes by host, protocol distribution, conversation statistics

Discovery

Wireshark revealed:

- Backup server → Cloud storage
- 500+ Mbps sustained transfer
- Scheduled backup job at 3 PM
- Runs during business hours!

Resolution

Reschedule backups to 2 AM; implement QoS for backup traffic; add bandwidth monitoring alerts

Key Lesson

Packet capture reveals what dashboards miss. **Statistics** → **Endpoints** is your friend!

Section 8.6: Traffic Monitoring & Flow Data

Flow Data vs Packet Capture

	Flow	**Packet**
Content	Metadata	Full data
Storage	Low	High
Privacy	Better	Sensitive
Detail	Summary	Complete
Use	Trends	Deep dive

Flow = Conversation Summary

Source IP, Dest IP, Ports, Protocol, Bytes, Packets, Duration, Timestamps

Flow Protocols

Protocol	**Vendor**	**Port**
NetFlow	Cisco	UDP 2055
sFlow	Multi	UDP 6343
IPFIX	Standard	UDP 4739
J-Flow	Juniper	UDP 2055

QoS and Traffic Shaping

QoS Purpose

Prioritize critical traffic when bandwidth is limited:

- Voice/video over bulk data
- Business apps over social media
- Guaranteed bandwidth for critical systems

DSCP Values (Review)

Class	DSCP	Use
EF	46	Voice
AF41	34	Video
AF21	18	Critical data
BE	0	Default

Policing vs Shaping

Policing: Drop excess traffic immediately (hard limit, can cause packet loss)

Shaping: Buffer and delay excess traffic (smoother flow, adds latency)

QoS Only Helps When...

There's **congestion**! QoS doesn't add bandwidth—it manages limited bandwidth intelligently.

Module 8 Summary and Key Takeaways

Documentation & Policies

- **Change management**: Request → Review → Approve → Implement
- **Physical diagrams**: Where things are
- **Logical diagrams**: How things connect
- **IPAM**: Track all IP allocations

Monitoring & Analysis

- **SNMP**: v3 for security (auth + encryption)
- **Syslog**: Levels 0–7 (Emergency to Debug)
- **Packet capture**: tcpdump, Wireshark
- **Flow data**: NetFlow, sFlow, IPFIX

Key Ports Reference

Protocol	Port	Purpose
SNMP	UDP 161	Agent queries
SNMP Trap	UDP 162	Alerts
Syslog	UDP 514	Logging
Syslog TLS	TCP 6514	Secure logs
NetFlow	UDP 2055	Flow export
sFlow	UDP 6343	Flow export

Exam Reminders

Syslog 0 = Emergency, 7 = Debug; SNMPv3 = auth + encryption; SPAN/mirror for packet capture